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Unit “ 4 : Session “ 9 : SLO “ 1 | Pipelining “ Choose the Correct Answer

■ I. Multiple Choice

1. b) Increase instruction throughput “ pipelining overlaps stages so more instructions complete per unit time.
2. b) 5 “ the classic RISC pipeline has 5 stages: Fetch, Decode, Execute, Memory, Write-back.
3. b) Data hazard “ an instruction needing the not-yet-written result of a previous instruction.
4. b) Branch prediction “ guesses the branch outcome to keep the pipeline full through control hazards.
5. c) Fetch “ the IF stage reads the instruction from memory.
6. c) Stall “ a pipeline bubble is a stall cycle where no useful work advances.
7. a) Speedup = Non-pipelined time / Pipelined time.
8. a) Cycles Per Instruction.
9. b) Bypass unit “ forwarding (bypassing) routes ALU results directly to where they are needed.
10. a) Longest pipeline stage “ the slowest stage fixes the clock period and hence caps the speedup.